Intent-Adaptive Hybrid Retrieval and Heterogeneous-Space Person Fusion for Conversational Video Surveillance

Sujal M H*, Sujnan Kumar*, Yashas Shetty*, Suhan D Shet*, and Ravinarayana B†

*Department of Computer Science & Engineering,
Mangalore Institute of Technology & Engineering, Moodabidre, India

†Head, Department of Computer Science & Engineering,
Mangalore Institute of Technology & Engineering, Moodabidre, India

Email: {sujalmh9, sujnankumar439, yashass825, suhanshet2004}@gmail.com, hodcse@mite.ac.in

***Abstract*—Hybrid retrieval over surveillance video typically combines a structured metadata filter with a dense semantic index using a *fixed* mixing weight, which catastrophically mis-ranks counting and action queries. We make three novel contributions that, in combination, raise mean Precision@5 (macro-averaged over five query categories) on a 150-query natural-language benchmark from 0.67 (best fixed-α baseline, α=0.3) to 0.83. (C1) An *intent-adaptive blending function* $\alpha(F, \iota, n_s, n_v)$ that produces a closed-form, per-query mixing weight from the parsed filter F, the predicted query intent ι, and the structured/semantic result-set imbalance, with two hard overrides for counting and action queries (Eq. (3)); we show (under empirical concavity assumptions) why fixed-α schemes cannot match the per-category optima simultaneously. (C2) A *heterogeneous-space person descriptor* that fuses a SigLIP visual embedding ($\mathbb{R}^{768}$) and a Sentence-Transformers attribute-text embedding ($\mathbb{R}^{384}$) into a single L2-normalized $\mathbb{R}^{1152}$ vector by weighted concatenation rather than the naive same-space addition used by prior multimodal Re-ID work; an ablation shows a 10.7-pp F_1 gain over the best unimodal index and a 2.2-pp gain over a same-space late-fusion variant. (C3) A *conversational alert grammar* that reuses the same structured-output LLM parser to materialize fully-typed alert rules (event, object, zone, time, severity, cooldown, side-effects) from free-text utterances, eliminating manual rule authoring; we report 91% rule-correctness on a held-out alert-authoring benchmark. The contributions are deployed inside a nine-model real-time pipeline that sustains 2 FPS per camera over a 4-camera deployment with a 109 ms median FAISS lookup over 312k indexed person descriptors.**

***Index Terms*—Hybrid Retrieval, Intent-Adaptive Ranking, Multimodal Person Re-ID, Heterogeneous Embedding Fusion, Conversational Alert Authoring, Vision–Language Surveillance, Natural Language Querying.**

I. Introduction

NATURAL-language interfaces over surveillance video are now technically feasible: instance segmentation, person Re-ID, and contrastive vision–language models are mature, and structured-output LLMs can convert a user utterance into a typed query schema. The unsolved problem is *how to rank* the resulting candidate frames: a structured filter (*"persons in Zone A between 22:00 and 23:00"*) and a semantic vector lookup (*"red jacket loitering near the entrance"*) produce two incommensurable score scales. The dominant fusion strategies in the literature are a fixed convex sum $\alpha S_{\text{struct}} + (1-\alpha) S_{\text{sem}}$ tuned once per system [4], [6], [7], [23], parameter-free Reciprocal Rank Fusion [22], or sequential filter-then-rank pipelines [24]. We show empirically (Table III) that no single α simultaneously serves counting, finding, tracking, and action queries: a fixed $\alpha = 0.5$ collapses to 0.18 P@5 on action queries and a fixed $\alpha = 1.0$ collapses to 0.04 P@5 on find queries, while no fixed value exceeds 0.67 macro-mean P@5 across all five categories.

Two further open problems compound this. The per-person descriptor in prior multimodal Re-ID work [17], [25]–[28] either fuses same-space embeddings by addition or trains a joint dual encoder, both requiring dimension-matched backbones or joint contrastive data that surveillance deployments rarely have. And alert authoring in commercial engines (Verkada Command, Avigilon Unity, BriefCam Investigator) requires an IT operator to translate a behavioral description into a typed rule, capping the number of useful rules per deployment.

Contributions

1) **Intent-adaptive hybrid retrieval** (§IV). A closed-form, training-free mixing function $\alpha(F, \iota, n_s, n_v)$ with two hard overrides; an override-gap argument (Proposition 1) under empirical concavity; +0.16 macro-mean P@5 over the best fixed-α baseline.
2) **Heterogeneous-space person fusion** (§V). A weighted-concatenation descriptor that fuses backbones of incompatible dimensionality without joint training; +10.7-pp F_1@10 over the strongest unimodal index, +2.2-pp over a same-space late-fusion baseline.
3) **Conversational alert authoring** (§VI). A typed grammar reusing the query parser; 91% full-rule correctness (all 9 fields including cooldown and cadence) on a 60-utterance held-out benchmark.
4) **An end-to-end open implementation** validating all three contributions on a 4-camera live deployment (§VIII).[1]

[1]Code: https://github.com/sujalmh/intent-adaptive-hybrid-surveillance (to be released upon acceptance).

TABLE I: Position relative to closest related work along the three novel axes; entries indicate the kind of mechanism used.

System / Work	Per-query adapt. α	Heterog.-sp. Re-ID fusion	Conv. alert authoring
Fixed-α hybrid [23]	fixed	—	—
RRF [22]	rank-only	rank-only	—
HyST [24]	sequential	—	—
Agentic RAG [2]	route-only	—	—
VIDEX [4]	✗	✗	✗
3-Stage DL [7]	✗	✗	✗
Object-IR [6]	✗	✗	✗
LLM-Surv. [1]	✗	✗	✗
CUHK-PEDES [25]	—	joint-trained	—
CLIP-ReID [26]	—	joint-trained	—
Img-Txt Fusion [27]	—	same-space	—
TriReID [28]	—	same-space	—
OSNet [17]	—	visual-only	—
SigLIP/CLIP [19]	—	same-space	—
This work	✓	✓	✓

II. Related Work and Position

We compare directly with the closest published work along the three contribution axes (Table I).

Hybrid structured/semantic retrieval. Combining dense and structured retrieval is established practice in IR [20], [22], [23], dominantly via fixed-α convex combination [23] or parameter-free RRF [22]. Query-dependent weighting in IR (learning-to-rank, query-dependent fusion) typically learns weights from large-scale click data on text-only sparse/dense pairs and lacks hard overrides for semantically infeasible branches. Recent analogues HyST [24] (sequential LLM-filter then dense) and Agentic-RAG [2] (router-based) still do not produce a per-query blended score; surveillance forensic-search systems (VIDEX [4], three-stage DL [7], object-IR [6]) and CLIP+RAG pipelines [1], [14] use a single retrieval branch. We are not aware of prior surveillance-video work that computes the mixing weight per-query from parsed intent and result-set evidence with category hard overrides.

Multimodal person Re-ID. Heterogeneous-dimensional concatenation of frozen encoders is a known retrieval pattern (e.g., RAG [3]); we do not claim it is new in IR. We claim its use as the *primary* surveillance-Re-ID descriptor: prior NL-based person-search [25]–[28] couples a dual encoder by joint training or by descriptive-fusion over same-dimensional modalities, while same-space late-fusion [22] requires dimension-matched encoders or two indices. We are not aware of a surveillance Re-ID system using weighted concatenation of heterogeneous frozen encoders as a single FAISS index calibrated on as few as 50 labeled pairs.

Conversational alert authoring. Existing surveillance products [4], [5], [7] expose form-driven rule editors and structured-output LLMs [15] are standard for slot-filling; the novelty is the specific reuse of the same query-parsing schema to materialize a 9-tuple typed alert rule end-to-end, rather than a new LLM capability.

III. Background and System Substrate

The novel contributions are deployed inside a nine-model real-time pipeline (Table II). Live RTSP / MJPEG streams (1080p, 2 FPS) and uploaded files are processed by YOLOv11m-seg [8], BoT-SORT [16] with OSNet $x1\_0$ [17] for MOT/Re-ID, OpenVINO PAR-crossroad-0238 [9] for seven multi-label attributes, and a CIE LAB histogram with CIEDE2000 [10] naming over a 25-class CSS3 palette. Two FAISS `IndexFlatIP` indices [20] hold a scene embedding ($\mathbb{R}^{512}$, OpenCLIP ViT-B/32 [13]) and our novel person descriptor ($\mathbb{R}^{1152}$, §V). MongoDB stores 13 compound-indexed metadata collections; a FastAPI backend exposes 30+ REST endpoints to a Next.js dashboard. Captions come from a tiered VLM pipeline (Qwen2-VL-2B [11] with ViT-GPT2 / zero-shot CLIP fallbacks [12]–[14]); the NL parser is a structured-output LLM (GPT-4o-mini [15] by default; configurable to local Ollama).

TABLE II: Off-the-shelf components and their roles. None are claimed as contributions.

Component	Role	Output
YOLOv11m-seg [8]	Detection + masks	per-frame
BoT-SORT [16]	MOT association	track IDs
OSNet $x1\_0$ [17]	Re-ID appearance	512-d
PAR-crossroad-0238 [9]	Person attributes	7-d (sigmoid)
OpenCLIP ViT-B/32 [13]	Scene embedding	512-d
SigLIP-base-p16 [19]	Person visual	768-d
MiniLM-L12-v2 [18]	Attribute text	384-d
Qwen2-VL-2B [11]	Captioning	text
GPT-4o-mini [15]	NL parsing	JSON

IV. C1: Intent-Adaptive Hybrid Retrieval

A. Problem

A natural-language query q is parsed into a structured filter F (a Pydantic `NLFilter` of 16 typed fields) and a semantic-text payload q_{sem}. The structured branch retrieves $R_s = \{r\}$ with scores $S_{\text{struct}}(r) \in [0, 1]$ from MongoDB; the semantic branch retrieves R_v with scores $S_{\text{sem}}(r) \in [-1, 1]$ from FAISS. The hybrid score is the convex combination

$$S_{\text{hyb}}(r) = \alpha\, S_{\text{struct}}(r) + (1-\alpha)\, S_{\text{sem}}(r), \quad \alpha \in [0,1]. \quad (1)$$

Existing systems pick a single α per deployment. We show this is fundamentally insufficient.

B. Failure of fixed-α schemes

We sweep $\alpha \in \{0.0, 0.3, 0.5, 0.7, 1.0\}$ on the 150-query benchmark (Table III). The collapse is structural, not a tuning artifact. *Counting* queries (*"how many cars in Zone A"*) demand an exact integer that only the structured branch produces; the semantic score is a continuous similarity in $[-1, 1]$ that cannot represent "equals 5," so any weight on the semantic branch dilutes a correct count, and P@5 falls from 0.94 at α=1.0 to 0.31 at α=0.0. *Action* queries (*"person running"*) have no corresponding field in the 16-field `NLFilter` schema, so the structured branch returns either an unfiltered scan or empty; weight on the structured branch therefore injects noise, and P@5 falls from 0.69 at α=0.0 to 0.00 at α=1.0. The two requirements push the optimum to opposite endpoints of $[0, 1]$; the best fixed macro-mean is only

TABLE III: Fixed-α baselines on the 150-query NL benchmark (30 queries per category). **Mean** is the macro-average across the five categories. No single value dominates; counting and action queries pull in opposite directions (boldface = best per row, italics = worst).

α	Count	Find	Track	Action	Filter	Mean
0.0 (semantic only)	*0.31*	0.74	0.79	**0.69**	0.55	0.616
0.3	0.48	0.76	0.80	0.58	0.71	**0.666**
0.5	0.69	0.71	0.72	*0.18*	0.84	0.628
0.7	0.86	0.62	0.61	0.07	0.88	0.608
1.0 (structured only)	**0.94**	*0.04*	*0.02*	*0.00*	**0.92**	*0.384*
Adaptive (Eq. (3))	**0.94**	**0.78**	**0.81**	**0.69**	**0.92**	**0.828**

0.666 at α=0.3, motivating a per-query mixing weight rather than a per-deployment one.

C. Adaptive blending function

Let $C : \mathcal{F} \to \{\textsc{cnt}, \textsc{act}, \textsc{soft}\}$ be the *category function* on parsed filters, defined as

$$C(F) = \begin{cases} \textsc{cnt}, & \text{if } F \text{ contains a numeric count constraint,} \\ \textsc{act}, & \text{else if } F \text{ contains an action-verb token,} \\ \textsc{soft}, & \text{otherwise.} \end{cases} \tag{2}$$

We then define

$$\alpha(F, \iota, n_s, n_v) = \begin{cases} 1.00, & C(F){=}\textsc{cnt}, \\ 0.10, & C(F){=}\textsc{act}, \\ \mathrm{clip}_{[0.10,0.95]}(\alpha_{\text{soft}}), & \text{else,} \end{cases} \tag{3}$$

$$\alpha_{\text{soft}} = 0.6\,\alpha_\iota + 0.2\,\rho_{\text{cplx}} + 0.2\,\rho_{\text{src}}, \tag{4}$$

where CNT (count constraint) and ACT (action verb) are the two override categories of $C(F)$. The intent prior is $\alpha_\iota \in \{0.9, 0.5, 0.7\}$ for $\iota \in \{\text{count}, \text{track}, \text{find}\}$, query complexity $\rho_{\text{cplx}} = \min(1, |F|/4)$, and structured-evidence ratio $\rho_{\text{src}} = n_s/(n_s + n_v + \varepsilon)$ where $n_s = |R_s|, n_v = |R_v|$, $\varepsilon = 10^{-6}$. The two hard overrides encode domain truths: a count constraint (e.g., *"how many cars"*) is unanswerable from semantic similarity, and an action verb (e.g., *"running"*) is unanswerable from the structured filter, which has no `action` field schema.

D. Why fixed α cannot match

Proposition 1 (Override gap, under empirical assumptions)**.** Let C_1, C_2 be two queries with conflicting overrides ($C(F_1) = \textsc{cnt}$, $C(F_2) = \textsc{act}$), and assume the empirical P@5 curves $a \mapsto \text{P@5}(a, C_i)$ are concave on $[0.1, 1]$ with maxima at $a = 1$ and $a = 0.1$ respectively (verified on Table III). For any fixed $\alpha^\star \in [0.1, 1]$, the worst-case suboptimality

$$\begin{aligned} \Delta(\alpha^\star) = \max\big[\, & \text{P@5}(1, C_1) - \text{P@5}(\alpha^\star, C_1), \\ & \text{P@5}(0.1, C_2) - \text{P@5}(\alpha^\star, C_2)\,\big] \end{aligned} \tag{5}$$

satisfies $\Delta(\alpha^\star) \geq \frac{1}{2}\big(P_{1,1} - P_{1,0.1}\big)$, where $P_{i,a}$ denotes the empirical P@5 of category C_i (first subscript: category index $i \in \{1, 2\}$) at mixing weight $\alpha = a$ (second subscript).

Argument. Any $\alpha^\star \in [0.1, 1]$ is at distance at least 0.45 from one of the two override points. Combined with the stated concavity, this distance lower-bounds the loss against the maximizer of one of the two categories. The argument is empirical, not a worst-case theoretical guarantee; if the assumed concavity is violated the bound need not hold. □

On our benchmark $P_{1,1} - P_{1,0.1} = 0.94 - 0.31 = 0.63$, so any fixed α loses ≥ 0.31 P@5 on at least one category; this matches the empirical 0.94 vs. 0.18 gap in Table III.

E. Soft-zone behavior

For non-override queries, α_{soft} interpolates intent prior, complexity, and per-branch evidence (weights 0.6/0.2/0.2); the clipping interval $[0.10, 0.95]$ prevents either branch from being silenced. Realized α ranges are reported in Table IV.

TABLE IV: Realized α ranges on the 150-query benchmark.

Query Class	Realized α	Effect
Count constraint	1.00	Pure structured (override)
Action verb	0.10	Semantic-dominant (override)
Count intent	0.74–0.86	Structured-leaning (soft)
Find intent	0.58–0.70	Balanced (soft)
Track intent	0.40–0.52	Semantic-leaning (soft)

F. Post-ranking

Top-k candidates are then re-ranked by an exponential recency boost $S' = S_{\text{hyb}} \cdot 2^{-\Delta t/h}$, h=24 h, and Maximal Marginal Relevance [21] with λ=0.3.

V. C2: HETEROGENEOUS-SPACE PERSON FUSION

A. Problem

Multimodal person Re-ID [25]–[28] typically requires either two same-dimensional encoders fused by addition, or a jointly trained dual encoder. In a deployment we must work with whatever pre-trained encoders are available, here SigLIP-base ($\mathbb{R}^{768}$) [19] and MiniLM-L12-v2 ($\mathbb{R}^{384}$) [18], whose dimensions do not match.

B. Construction

For each person ROI we form

$$\begin{aligned} \mathbf{v} &= \text{SigLIP}(\text{Mask}(\text{ROI})) \in \mathbb{R}^{768}, \\ \mathbf{t} &= \text{MiniLM}\big(\text{Text}(\hat{y}, C)\big) \in \mathbb{R}^{384}, \end{aligned} \tag{6}$$

where $\text{Text}(\hat{y}, C)$ is the natural-language sentence assembled from the seven sigmoid attribute predictions $\hat{y}$ and the named color C (e.g., *"person wearing red coat, carrying bag, long sleeves, male"*). The fused descriptor is

$$\mathbf{e}_{\text{fused}} = \frac{[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]}{\big\|[\,\alpha_v\,\mathbf{v} \parallel \beta_t\,\mathbf{t}\,]\big\|_2} \in \mathbb{R}^{1152}, \tag{7}$$

with $(\alpha_v, \beta_t) = (0.6, 0.4)$ obtained by coarse grid search on the unit simplex (step 0.1, $\alpha_v + \beta_t = 1$, nine pairs) maximizing F_1@10 on a held-out 50-pair calibration set. The single L2 normalization makes the inner product a valid cosine similarity searchable by FAISS `IndexFlatIP`.

TABLE V: Person-index ablation. "Add (proj.)" projects $\mathbf{t}$ to $\mathbb{R}^{768}$ via a random Gaussian projection then adds; "Late (RRF)" performs Reciprocal Rank Fusion [22] on two parallel indices; "Concat (ours)" is Eq. (7).

Person index	P@5	R@10	F_1@10
Visual only ($\mathbf{v}$, $\mathbb{R}^{768}$)	0.712	0.689	0.700
Text only ($\mathbf{t}$, $\mathbb{R}^{384}$)	0.645	0.612	0.628
Add (proj. $\mathbf{v}+\mathbf{Pt}$, $\mathbb{R}^{768}$)	0.768	0.741	0.754
Late fusion (RRF, two indices)	0.801	0.770	0.785
Concat (ours), $(0.6, 0.4)$**,** $\mathbb{R}^{1152}$	**0.823**	**0.791**	**0.807**

C. Why concatenation, not addition

Same-space addition $\mathbf{v}+\mathbf{t}$ requires a projection $\mathbf{P} \in \mathbb{R}^{768\times 384}$, is sensitive to relative norm, and collapses two information sources into one coordinate per dimension. Weighted concatenation preserves both encoders' full rank at the cost of one normalization, requires no joint training, and the only learned parameter is the scalar pair (α_v, β_t).

D. Ablation

The concat construction beats the strongest unimodal baseline by 10.7 pp F_1@10 and the strongest fusion baseline (RRF) by 2.2 pp, using one FAISS index instead of two. A coarse (α_v, β_t) sweep is monotonic around $(0.6, 0.4)$: $(0.5, 0.5)$=0.785, $(0.7, 0.3)$=0.799, $(0.4, 0.6)$=0.774 F_1@10.

E. Indexing complexity

Adding a person is $O(d) = O(1152)$; querying k neighbors over N persons with `IndexFlatIP` is $O(Nd)$. On our hardware this yields a 109 ms median for $N = 312{,}544$ descriptors (Table XI); the corresponding scene-index lookup mean is 107 ms.

VI. C3: Conversational Alert Authoring

A. Problem

Commercial alert engines [4], [5], [7] require an operator to translate a behavioral intention into a typed rule (event family, object class, zone, threshold, time window, severity, cooldown, side-effects). For non-technical operators this is a productivity bottleneck that limits the number of useful rules in a deployment.

B. Grammar

We define a typed alert rule as a 9-tuple

$$\mathcal{R} = \langle\, e,\, \mathcal{O},\, \zeta,\, \tau,\, \mathcal{T},\, s,\, c,\, \Sigma,\, \kappa \,\rangle$$

with $e \in \{\text{crowd}, \text{loiter}, \text{fight}, \text{zone}, \text{count}, \text{time}\}$, $\mathcal{O} \subseteq \mathcal{C}_{\text{det}}$ a subset of detected classes, ζ a zone label, $\tau \in \mathbb{R}_{\geq 0}$ a numeric threshold, $\mathcal{T} = [t_1, t_2]$ a time window with cross-midnight support, $s \in \{\text{low}, \text{med}, \text{high}\}$, $c \in \mathbb{N}$ the cooldown in seconds, $\Sigma \subseteq \{\text{store\_clip}, \text{snapshot}, \text{push\_ws}\}$ side-effects, and $\kappa \in \mathbb{R}_{>0}$ an evaluation cadence (Hz). The grammar is materialized as a Pydantic schema; the same structured-output LLM used for queries fills it.

TABLE VI: Rule families and triggers.

Rule Family	Trigger Condition
Crowd density	$\#\{\text{persons}\} \geq \tau_{\text{crowd}}$ within camera/zone
Loitering	$\Delta t_{\text{track}} \geq 60$ s for any track
Fight detection	$\bar{\mathcal{E}}_W \geq 0.03 \wedge \exists i \neq j : \lVert c_i - c_j \rVert < 24$ px
Zone occupancy	$\#\{\text{persons in ROI}\} >$ capacity
Count rules	$\#\{c\} \,\square\, n,\ \square \in \{=, \geq, \leq, >, <\}$
Time-of-day	$t \in [t_{\text{start}}, t_{\text{end}}]$ (cross-midnight)

TABLE VII: Z-score anomaly detector (7-day labeled subset).

Event Type	Precision	Recall	F_1	FPR/day
`crowd_spike`	0.872	0.811	0.840	1.4
`off_hours`	0.804	0.866	0.834	2.1
`unusual_object`	0.713	0.792	0.751	3.7

C. Authoring example

The utterance *"alert me when more than five people gather in Zone A after 10 PM"* produces

$$\begin{aligned}\mathcal{R} = \langle\, &\text{crowd},\ \{\text{person}\},\ \text{Zone A},\ 5,\ [22{:}00, 06{:}00],\\ &\text{med},\ 60,\ \{\text{snapshot}, \text{push\_ws}\},\ 1\text{Hz}\,\rangle\end{aligned}$$

without any manual configuration. Six rule families are evaluated per-frame against the live detection stream (Table VI); each rule is cached from MongoDB with a 5-second TTL.

D. Complementary statistical anomaly detector

To surface unknown behaviors we maintain a rolling 7-day baseline of detection counts $\{x_{c,h}^{(d)}\}_{d=1}^{7}$ per (camera, hour) bucket and emit an event when

$$Z_{c,h} = \frac{x_{c,h}^{*} - \mu_{c,h}}{\sigma_{c,h} + \varepsilon} \geq \tau_Z, \tag{8}$$

with $\tau_Z \in \{1.5, 2.0, 3.0\}$ for unusual-object, off-hours, and crowd-spike events (Table VII).

VII. Implementation Detail (Brief)

The complete detection-and-indexing loop is given in Algorithm 1; the resolution loop that consumes Eqs. (3) and (7) is Algorithm 2. The end-to-end pipeline is organized as five dataflow layers (capture, perception, indexing, reasoning, output) in which the three novel components (C1, C2, C3) are the only non-off-the-shelf blocks; all other modules are listed in Table II.

Algorithm 1 Detection and indexing per frame.

Input: frame F_t, timestamp t
$D_t \leftarrow \text{YOLOv11mSeg}(F_t, \tau_{\text{det}}{=}0.5)$
for each $(b_i, m_i, c_i, p_i) \in D_t$ **do**
  $\mathbf{q} \leftarrow \text{LABHistogramMode}(F_t[b_i], m_i)$
  $C \leftarrow \arg\min_{\mathbf{p}_j} \Delta E_{00}(\mathbf{q}, \mathbf{p}_j)$
  if $c_i{=}\texttt{person} \wedge w(b_i) \geq 60$ **then**
    $\hat{y} \leftarrow \sigma(\text{PAR}(\text{Letterbox}(\text{Mask}, 80, 160)))$
    $\mathbf{e} \leftarrow$ Eq. (7); FAISS$_p$.add($\mathbf{e}$)
  end if
  MongoDB.insert(record)
end for
if $t \bmod \Delta_{\text{idx}} = 0$ **then**
  EnqueueSemanticIndexing(F_t)
end if

Algorithm 2 NL query resolution using Eqs. (3), (7).

Input: NL query q
$F, q_{\text{sem}}, \iota \leftarrow$ LLM_Parse(q)
$R_s \leftarrow$ MongoDB.find(F); $R_v \leftarrow$ FAISS.search($g_T(q_{\text{sem}}), k$) $\{g_T$ is the OpenCLIP ViT-B/32 text encoder paired with the scene index$\}$
$\alpha \leftarrow \alpha(F, \iota, |R_s|, |R_v|)$ {Eq. (3)}
$R \leftarrow \text{Merge}(R_s, R_v)$; rescore by Eq. (1)
$R \leftarrow \text{RecencyBoost} \circ \text{MMR} \circ \text{ConfFilter}(R)$
return $R[: k]$, LLM_Answer(q, R)

VIII. Evaluation

A. Setup

Hardware: Intel i7-13700K, RTX 4070 (12 GB), 32 GB DDR5, Windows 11. Software: Python 3.12, PyTorch 2.3, CUDA 12.1, OpenVINO 2024.2, FAISS 1.8, MongoDB 7. Live evaluation: four 1080p IP cameras at 2 FPS for $\sim$8 h ($\sim$230k frames, 1.25M detections). Retrieval evaluation: a held-out 150-query NL benchmark spanning five categories (30 queries each) with human-annotated relevant clip sets, plus a 60-utterance held-out alert-authoring benchmark with reference rule annotations.

Statistical protocol. We report mean P@5 with a 95% bootstrap CI (10,000 query-level resamples). Paired comparisons (adaptive vs. fixed-α, concat vs. each fusion baseline) use a two-sided paired bootstrap on per-query P@5 deltas plus a Wilcoxon signed-rank cross-check; significance threshold $p < 0.05$.

B. C1 – adaptive-α vs. fixed-α

Adaptive-α raises macro-mean P@5 from the best fixed-α value $0.666\,(\pm0.039)$ at $\alpha{=}0.3$ to $0.828\,(\pm0.034)$, an absolute gain of $+0.162$ with paired-bootstrap 95% CI $[+0.118, +0.207]$ over 150 queries (Table VIII). The paired bootstrap and Wilcoxon signed-rank tests both reject equality at $p < 10^{-4}$; per-category gains against the per-category best fixed-α are zero on the override categories (count, action) and on filter (where $\alpha{=}1.0$ already saturates), and positive on find ($+0.02$, $p = 0.04$) and track ($+0.01$, $p = 0.07$). Crucially, the gain in macro-mean comes from the fact that no single fixed α can match the count, action, and filter optima simultaneously (Proposition 1); adaptive-α does so by construction.

TABLE VIII: Per-category P@5 with 95% bootstrap CIs (10,000 resamples, $n{=}30$ per category, $n{=}150$ overall). Δ is the per-query mean of (adaptive − per-category best-fixed-α); p is two-sided paired bootstrap. The bottom row compares the best single fixed-α macro-mean ($\alpha{=}0.3$) against adaptive-α.

Category	Best fixed-α	Adaptive	Δ (p)
Count	0.94 ± 0.05 ($\alpha{=}1.0$)	0.94 ± 0.05	$+0.00$ (n.s.)
Find	0.76 ± 0.07 ($\alpha{=}0.3$)	0.78 ± 0.07	$+0.02$ (0.04)
Track	0.80 ± 0.06 ($\alpha{=}0.3$)	0.81 ± 0.06	$+0.01$ (0.07)
Action	0.69 ± 0.08 ($\alpha{=}0.0$)	0.69 ± 0.08	$+0.00$ (n.s.)
Filter	0.92 ± 0.05 ($\alpha{=}1.0$)	0.92 ± 0.05	$+0.00$ (n.s.)
Macro-mean (best single fixed-α)	0.666 ± 0.039	$\mathbf{0.828 \pm 0.034}$	$\mathbf{+0.162}$ ($< 10^{-4}$)

TABLE IX: Conversational alert authoring on a held-out 60-utterance set. "Field-level" = per-field exact match, "Rule-correctness" = all 9 fields correct.

Field	Acc.	Hardest mistake
Event family	0.97	loiter vs. zone
Object set	0.95	"vehicle" $\to$ {car, truck}
Zone	0.88	fuzzy zone names
Threshold τ	0.93	implicit ("a few")
Time window	0.92	cross-midnight
Severity	1.00	—
Cooldown	0.98	—
Side-effects	0.95	implied snapshot
Cadence	1.00	—
Rule-correctness	**0.91**	—

C. C2 – fusion ablation

The heterogeneous-space concat descriptor reaches $\text{F}_1@10 = 0.807\,(\pm0.024)$, beating the best unimodal index (0.700 ± 0.031) by $+10.7$ pp (paired bootstrap 95% CI $[+7.9, +13.4]$, $p < 10^{-4}$) and the best fusion baseline RRF (0.785 ± 0.026) by $+2.2$ pp (95% CI $[+0.4, +4.1]$, $p = 0.018$, Wilcoxon $p = 0.022$). The same-space additive baseline (random Gaussian projection) gives $+5.3$ pp over visual-only ($p = 0.003$) but remains 5.3 pp below the concat construction ($p = 0.001$). Significance therefore holds against both unimodal and fusion baselines, with the concat-vs-RRF margin being the smallest and least conservative.

D. C3 — alert authoring

The 60 held-out utterances exercise all nine rule fields (Table IX); "Rule-correctness" requires all nine to be correct. Closed-vocabulary fields (severity, cadence, cooldown, event family) reach 0.97–1.00. The hardest are zone naming (0.88, fuzzy references such as *"the back entrance"* vs. canonical `Zone B`) and time window (0.92, cross-midnight intervals such as *"after 10 PM"* parsed as $[22{:}00, 06{:}00]$); threshold (0.93) misses on implicit quantifiers (*"a few"*), and object-set expansion (0.95) under-expands category nouns (*"vehicle"* $\to$ $\{\texttt{car}\}$). Aggregated, 0.91 of utterances yield a fully-correct rule on the first attempt; the remaining 9% require a single zone- or threshold-level correction, never a structural rewrite.

TABLE X: Substrate metrics (YOLOv11m-seg detection, BoT-SORT+OSNet tracking mean over 4 cameras, PAR macro, color naming on 1,568 ROIs, Z-score anomaly).

Stage	Metric	Value
Detection (person)	Precision / F_1 / mAP@0.5	0.965 / 0.927 / 0.873
Tracking (mean)	MOTA / IDF1	0.909 / 0.865
Person attributes	Macro F_1 (7 attrs)	0.882
Color naming	CIEDE2000 / RGB-Euclid	0.923 / 0.821
NL parsing	Field-correctness	0.94

TABLE XI: End-to-end query latency by category. "Sem. only" isolates the FAISS lookup; "Total" includes LLM parsing, hybrid blending, MMR, and clip stitching. Person index has $N{=}312{,}544$ descriptors; person-only median is 109 ms.

Category	Parse (ms)	Sem. only (ms)	Total (ms)	P@5
Count	1,768	94	3,441	0.94
Track	2,101	109	5,413	0.81
Find	6,405	121	11,983	0.78
Action	2,423	137	6,131	0.69
Filter	1,590	82	3,108	0.92
Mean	2,857	**107**	6,015	**0.83**

E. Supporting numbers

Substrate metrics are summarized in Table X; the Z-score anomaly detector (Eq. (8)) is in Table VII.

The 4-camera deployment sustains 2.0 FPS per stream at 4.2 GB resident memory, 7.8 GB VRAM, and 68% mean GPU utilization (91% p95) on the RTX 4070.

IX. Discussion and Conclusion

Generalization. All three contributions are schema-decoupled: C1 needs only a category function $C(\cdot)$ with finitely many overrides; C2 admits any pair of frozen encoders, with only (α_v, β_t) learned on a 50-pair calibration set; C3's grammar requires only a typed-schema and trigger-function extension per new event family.

Limitations. (i) Proposition 1 is empirical: it assumes per-category concavity of P@5 in α, verified on our benchmark but not guaranteed in general. (ii) Concat fusion costs $d_v + d_t$ bytes per descriptor; galleries with $N > 10^7$ would motivate IVF-PQ over flat IP. (iii) Alert-grammar correctness depends on the LLM; we observed a 6-pp drop with local 7B models. (iv) *Demographic bias and privacy.* PAR-crossroad-0238 was trained on a corpus skewed toward adult, daylight imagery; its binary *gender* attribute is a coarse visual proxy. We mitigate by using gender only as a soft cue, providing a switch to disable it, and applying a per-camera retention cap on raw imagery.

Conclusion. The three contributions raise macro-mean P@5 from 0.67 to 0.83, gain 10.7-/2.2-pp F_1@10 over unimodal/RRF baselines, and reach 91% full-rule alert-authoring correctness on a 4-camera, 312k-descriptor deployment, showing that schema-aware fusion closes the gap between dense retrieval and a usable conversational surveillance interface.

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